Subject Code: CS23532.

INDEX

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STD. _____ DIV. _____ ROLL NO. 240701317 SCHOOL _____

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Exp. No: 1 Study of Networks commands.

20/2/26

Windows commands

1-) arp -a : ARP is short form of Address Resolution Protocal.

O/P:

Interface: 10.16.50.56...0x9.

Internet Address Physical Address Type

2-) Lostname:

It will display the name of your computer

O/P: LAPTOP-T003BKTN

3) ipconfig /all : this command display configuration information about your TCP/IP connection.

O/P:

HostName LAPTOP-T003BKTN

Node type . . . . . Mixed.

4) nbtstat -a : This command helps solve problems with netbios name resolution.

O/P:

Ethernet 2:

Node IpAddress: [192.168.56.1] ScopeId: []

Host not found

5.) netstat : This command display a verify of statistics about a computer active TCP/IP connection

o/p: Active connections.

Proto	Local Address	Foreign Address	State
TCP	10.16.56.50:49444	4.213.25.242:https	Established

6) nslookup : It display the DNS details such as IP. address of a particular computer.

o/p:
server : Unknown.
Address : 10.16.56.155

7) pathping : pathping is a unique command in windows and its basically a combination of the ping and tracer command

o/p:
cmd : pathping -g google.com.
Tracing route to google.com [142.250.66.142]

8.) Ping : ping command is the best way to check connectivity between two nodes.

o/p:
cmd : ping google.com
ping.

9.) Route : route is used to show/manipulate the IP routing table

o/p:

cmd : route

Linux Commands

1) IP commands:

(i) ip address show : display all Ip address in network interfaces

o/p: eth0 : inet 192.16 1.10/24.

(ii) ip address add : add an Ip address to an interfaces

o/p: cmd ip address add 192.168.1.254/24

IP address added successfully

(iii) ip link set ethernp : Enables the network interface

o/p: eth0 state up.

(iv) ip link set eth0 down : Disables the network interface

o/p: eth0 state down.

(v) ip link set eth0 promise on : Enables permissions mode

o/p: promiscous mode enables.

(vi) ip route add default via 192.168.1.154 dev eth0: Adds the default gateway

vii) ip route get 10.10.1.4 : shows the route to the destinations IP.

O/P : 10.10.1.4 via 192.168.1.254 dev eth0

2) ip config : displays network interfaces configuration

O/P : eth0 link en : ethernet inet 192.168.1.10

3) mtr :

(i) mtr google.com : shows route statistics to google.com

O/P : Host : localhost loss % avg RCT

(ii) mtr -b google.com : shows both hostname and IP address

O/P : google.com (142.250.183.206)

(iii) mtr -g google.com : shows numbers IP address only

O/P : 142.250.183.286.

(iv) mtr -c 10 google.com : sends 10 probe packets

O/P : start 10 packets.

4) tcpdump :

(i) dnf install -y tcpdump : install tcpdump

(ii) tcpdump -D : lists available capture interface

O/P : eth0 2.10.

(iii) tcpdump -i eth0 -c 10 host 8.8.8.8 capture traffic from a specific host to itself

O/P : 10 packets captured

(iv) tcp-dump -i etho dst host 8.8.8.8: capture traffic from a host

o/p: 8.8.8.8 > 192.168.1.10

(v) tcpdump -i etho dst host 8.8.8.8: capture traffic going to a host

O/p: 192.168.1.10 > 8.8.8.8

(vi) tcpdump -i etho port 53: capture DNS port 53 traffic.

o/p: DNS query captured.

(vii) tcpdumpcd -i etho -c 10 host www.google.com and port 443: capture only HTTPS traffic

o/p: TCP 443 captured.

(viii) tcpdump -i etho port not 53 and not 25: captured all port traffic except ports 80 and 25.

Result:

The above both the windows and linux commands are executed and output are completed successfully.

28/2/26

Exp.No: 2

Date: 27/7/26

NETWORK CABLES.

AIM:

Study of different types of networks cables.

(a) Understand different types of Network cables

1. Unshield twisted pair cable
2. Shield twisted pair cable
3. Coaxial cable
4. Fibre optic cable.

Cable type	category	Maximum data transmission	Advantages / disadvantage	Application
UTP	category 3	10 bps	Advantage * Cheaper in cost	10 Base-tethernet
	category 5	upto 100 Mbps.	* Easy to install	Fast ethernet Gigabit ethernet.
	category	1 Gbps	Disadvantage * More prone to (EMI) Electromagnetic interfacce and noise	Fast ethernet, Gigabit Ethernet.

Cable type	Category	Maximum data transmission	advantage / Disadvantage	Application
STP	category 6, 6a	10Gbps	Advantage: * shielded * Faster than UTTP * Less susceptible to noise and interface Disadvantage: *Expensive * Ca	Gigabit ethernet. 10G ethernet (55 m) widely used in data centres
STTP.	category 7.			Gigabit ethernet, 10G. Ethernet (100 m).
coaxial	RG-6 RG-59 RG-11	10 - 100 Mbps.	* High bandwidth *Immune to interference *Low loss bandwidth. * Versatile * Disadvantage: *Limited distance * cost * size is Bulky	Speed of signal is 500m Television network High speed internet connection.

fibre optic Cable.	Single mode Multi mode	100 Gbps.	Advantage: * High speed * High bandwidth * High Security. * Long distance Disadvantage: * Expensive * Required skilled installers.	Maximum distance of fibre optics Cable is around 100 meters.

Result:

Different types of network cables are learned and studied successfully

Exp. No: 03
Date: 03/8/26

CISCO PACKET TRACER.

AIM:

To study the Cisco Packet tracer simulation tool understand it's user interface, design a simple network and analyze the behaviour of hub, switch and peer to peer

CISCO PACKET TRACER:

Cisco Packet tracer is a network simulation and modelling tool. it allows us to create and test computer network without using real networking equipment.

We can add PCs, switches, hubs, routers and other devices and observe how packets travel between them.

1. PEER TO PEER:

* Two computer communicate directly
* A Peer to peer network was created by connecting 2 PCs directly.

IP address were assigned to both PCs in the same network.

The connectivity between the 2PCs. was tested using the ping command that the two PC's could communicate directly with each other.

2. Network using HUB:

* Connected multiple computer and forwards received data to all ports.

* A Network was created by connecting 3 PCs to Hub using copper straight through cables. IP address were assigned through cables. IP address were assigned to all the PCs when data was sent from one PC to another, thus the HUB forward the packet to all connected ports. the distination PC received the packet and successfully communication was observed.

3. Network Switch :

* Connects computer and forward data toward the required destination.

* A network was a switch using copper straight cables. IP address were addressed were assigned to all PCs. and the connectivity was tested using the ping command. the switch forwarded the packet to the appropriate destination port allowing successful communication between the PCs.

4. Hub and switch :

Both Hub and switch were used in the network. the packet forwarding behaviour of both devices was observed and compared.

Result :

The peer to peer, Hub and switch were successfully designed and communication was established.

Exp. NO: 4

Date: 10/8/26

Setup and configure a LAN (Local Area Network) using a switch and ethernet cables in your lab.

AIM:

To setup and configure a LAN using a switch and ethernet cables in your lab.

What is LAN?

LAN stands for Local Area Network which is a group of computers and connected devices linked together in a small physical location like a school, home or office building.

How TO SET UP LAN?

STEP 1: Connect the hardware

* Place the router in the central location near in the central location near a power outlet.

* Connect your modem to the router's WAN or Internet port using a ethernet cable.

STEP 2: Connect your devices

* Wired setup: plug one end of ethernet cable into your computer network port and the numbered LAN.

Port on the router

Step 3 : Configure the router.

* Type your IP address 192.168.1.1 into the address bar and press Enter.

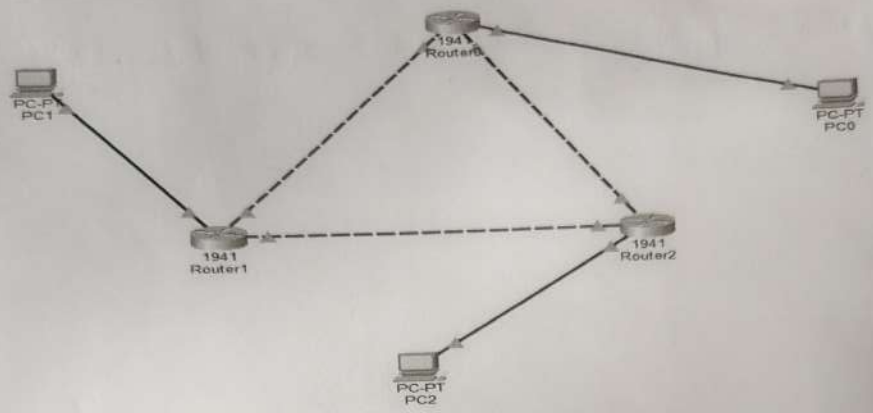
* Log in using the default username and password printed on the router.

Step 4 : Check the connectivity

Check the connectivity between switch and other machine by using PING COMMAND into the command prompt of the device.

Step 5: confirm the connectivity

Try to access the shared folder from other computers of the networks.



Result: Wireless LAN using switch and connect ethernet cables are executed successfully in cisco packet tracer.

Exp.No: 05
Date: 18/08/26

WIRESHARK

AIM:

To filter, capture, view, packets in Wireshark tool.

Procedure:

* select LOCAL AREA CONNECTION in Wireshakes.
* To capture option.
* select stop capture automatically after 100 packets.
* Then click shark capture
* Search TCP packets in search bar.
* Save the packets.

1. Create a filter to display in only TCP/UDP packets, inspects the packet and provide the flow graph.

* select LAN connections in Wireshark
* To capture option.
* select stop capture automatically after 100 packets.
* Then click start capture.
* Save the packets.

2. Create a filter to display only ARP pack and inspect the packet.

* Go to capture option.
* Select stop capature automatically after 100 packets.
* click shark capture.
* Search ARP Packets in search bar.
* save the packets.

3. Create a filter to display only HTTP packets and inspects the packet.

* Select local area connections in wireshark
* Go to capture option.
* Select stop capture automatically after 100 packets.
* Then click shark capture
* Search HTTP packets in search bar

4. create a file to display only DNS. packets and provide the flow graph.

* Go to capature option.
* select stop capture automatically after 100 packets.
* click start capture
* Search DNS Packets in search bar
* To use the flow graph click statistics flow graph.

5. Create a filter to display only DHCP Packets and inspect the packets.

* select stop capture automatically after 100 packets.
* the click start capture.
* search DHCP packets in search bar
* Save the packets.

Result:

Wireshark on packet capture tools experiment are studied and executed successfully.

EXP NO: 06

Date: 07/9/24

ERROR DETECTION AND CORRECTION USING HAMMING CODE.

AIM:

Write a program. To implement error detection and correction using hamming code concept.

Make a test run to input data stream and verify error correction feature.

SENDER:

* Input to sender file should be a text of any length program should be convert the text to binary.

* Apply hamming code concept on the binary data and add redundant bits to it.

Code:

```
data = input ("Enter a data:")
m    = len(data)
r = 0
while 2**r < (m+r+1):
    r+=1
hamming = [0]*(1, m+r+1)
j = 0
for (i in range 1, m+r+):
if (i-1)! =0.
```

```
hamming[i] = int(data[j])
j += 1
for i in range(3):
    pos = 2**i
    parity = 0
    for j in range(1, m+r+1):
        if j and pos != 0
            parity = hamming[]
```

RECEIVER:

Receiver program should read the input from channel file.

Apply hamming code on the binary data to check for error.

code:

```
Receiver = input("Enter the received hamming code")

n = len(receiver)
hamming = [0]*(n+1)
hamming[i] = int(receiver[n-i])
i=0
while 2**r < n+1:
    r += 1
    error=0
```

```
for i in range(r):
    pos = 2**
    Parity = 0
    for j in range(1, n+1):
        if j and pos != 0
            Parity = hamming[j]
    error += parity * pos:
if error != 0
    printf("Error Detected ; error)
else:
    Printf ("There is no error")

for i in range(n, 0, -1):
    Print (hamming[i], end = '')
Printf()
```

Sample Input:

Data bits : 1011

Hamming code : 1100110

Receive detected at position : 6

Correct hamming code : 1100110

Result:

Error detection and error connection using hamming code and learned and executed successfully

Exp.No: 7 Nmap to Discover live Hosts using Nmap scans on the tryHackme platform
Date: 22/9/26

Aim:

To discover live hosts using Nmap scans (ARP, ICMP TCP/UDP) on the tryHackme platform room link;

http://tryhackme.com/room/nmapol,

Procedure:

TASK 1

* ARP scan: This scan uses ARP request to discover live hosts.

* ICMP scan: This scan uses ICMP request to identify live hosts.

* TCP/UDP live PING scan: This scan sends packets to TCP ports and UDP ports to determine live hosts.

TASK 2 SUBNETWORK

Network segment: It is a collection of computer linked together through a common "medium".

subnetwork : It is usually the equivalent of one or more segments linked together and configured to utilize the same router in an IP network :

* Subnets with /16
* Subnets with /24.

Task 3 – ENUMERAING TARGETS.

We need to identify the targets we want to scans before we conduct any "actives"

1. List

< machine - IP > < scanme.nmap.org >

< example > -> will scans 3 IP address

2. Range

⇒ 10.11.12.15 - 20

⇒ will scan 6 IP address.

3. Subnet

⇒ < machine - IP > /30

⇒ Will scan 4 IP address

Result:

using TRYHACKME platform, we learned and executed to discover live hosts on NMAP scans (ARP, ICMP, TCP/CDP).

Exp.NO: 6
Date: 28/9/24

FLOW CONTROL USING SLIDING WINDOW PROTOCAL

Aim:

Write a program to implement flow control od data link layer using SLIDING WINDOW PROTOCAL

SENDER:

* Get the window size from User
* Get the text message
* Divide the message into frames
* Send to SENDER-BUFFER
* Wait for Acknowledgement
* Read receiver Buffer:
* If ACK is received, send the next
* If NACK is received, retransmit

RECEIVER:

* Read frames from SENDER. BUFFER
* check the frames number
* If the frames number is expected, send ACK
* otherwise send NACK
* Write ACK/NACK into

CODE :

SENDER

```
import time
w = int (input ("Window size : "))
msg = input ("Message : ")
i = 0
while i < len (msg)
    end = min (i + w, len (msg))
    with open ("SENDER_BUFFER", "w") as f:
        for j in range (i, end):
            f.write (f"{j} : {msg [j]} \n")
        Print ("frame", j, "→", msg [j])
    time.sleep (2)
    With open ("RECEIVER_BUFFER") as f:
        ack = f.read ()
    i = int (ack [i])
Printf ("Transmission completed")
```

RECEIVER

```
import time
import os
expected = 0
while true :
    if not os.path.exists ("sender_Buffer")
        continue
    with open ("SENDER_BUFFER") as f:
        frames = f.readlines ()
```

```
for x in frames:
    n = int(x.split(":")[0])
    if n == expected:
        expected += 1
        print("Received frame", n)
        with open("Receiver Buffer", "w") as f:
            f.write("ACK" + str(expected))
    else:
        with open("RECEIVER_BUFFER", "w") as f:
            f.write("NACK:" + str(expected))
        break
    time.sleep(1)
```

Result:

The SLIDING WINDOW PROTOCAL was implement successfully. ACK/NACK was simulated successfully.

Exp. No: 9
Date: 28/9

Implementation of subnetting in cisco packet tracer simulation

Aim:

To implement classless IP subnetting using cisco packet tracer and verify connectivity between different subnets

Procedure:

1. Open cisco packet tracer and create a new network topology.
2. Add routers, switches and pcs to the workspace.
3. connect the devices using appropriate cables.
4. Divide the network 192.168.1.0/24 into smaller subnets using the /27 subnets mask.
5. configure the IP address and subnet mask for the router interfaces.
6. configure the PC's with suitable IP address, subnet masks and default gateways.
7. configure the switch ports and routers interfaces.
8. use the PING command to test connectivity between the devices.
9. Verify that the packets are successfully transmitted between the connected networks.

Network	Subnet mask	Router IP	PC IP Range
192.168.1.0/27	255.255.255.224	192.168.1.1	192.168.1.11.15
192.168.2.0/27	255.255.255.224	192.168.2.1	192.168.2.11.15
192.168.3.0/27	255.255.255.224	192.168.3.1	192.168.3.11.15
192.168.4.0/27	255.255.255.224	192.168.4.1	192.168.3.11.15.

Result :

The classless IP subnetting was successfully implemented using cisco packet tracer.

Exp.No:10 END TO END ...

Date:22/9 AT TRANSPORT LAYER.

(a) Implement Echo client server using TCP sockets.

Aim:

To implement an ECHO SERVER applications using TCP sockets.

Code :

ECHO SERVER

```
import socket
s = socket.socket()
s.bind(("localhost", 5000))
s.listen()
c, addr = s.accept()
msg = c.recv(1024).decode()
print("client : ", msg)
c.send(msg.encode())
c.close()
s.close()
```

ECHO CLINT

```
import socket
s = socket.socket()
s.connect(("local host", 5000))
msg = Input("Enter message: ")
s.send(msg.encode())
print("Echo:", s.recv(1024).decode())
s.close()
```

Input:

Enter message: Hello server.

Output:

Server:

client connected

Received: Hello server

client

Echo from server: Hello server.

Result:

The Echo client server using TCP sockets was studied and executed successfully.

(6) Implement chat client server using TCP/IP sockets

Aim:
To implement a chat client server application using TCP sockets

CODE:

SERVER

```
import socket
s = socket.socket()
s.bind(("local host",5000))
s.listen(1)
c,a = s.accept()
while True:
    msg = c.recv(1024).decode()
    print("client :", msg)
    if msg == "bytle":
        break
    reply = input("server: ")
    c.send(reply.encode())
c.close()
```

CLINT

```
import socket
s = socket.socket()
s.connect(("localhost", 5000))

while True:
    msg = input("client, ())
    s.send(msg.encode())
    if msg == "bye":
        break
    Print("server:") s.recv(1024).
                    decode())

So close()
```

INPUT

client : Hello

server : Hi

client : Bye

OUTPUT

client : Hello

Server : Hi

Result:

The chat client server applications were successfully studied and implement using TCP socket.

Exp. No: 11
Date: 28/9 Implement your own ping program...

Aim:
To implement a ping programs using python to check the connectivity between source and destination host.

Algorithm:
1. Get the destination IP address from user
2. Send the request to destination
3. Wait for response
4. Calculate the response time
5. Display whether the host is reachable
6. Repeat the process for few request.

Code:

```
import os
import time
host = input ("Enter host :")
for i in range (4):
    start = time.time ()
    result = os.system ("ping -n " + host + "
                        null")
    end = time.time()
    if result == 0:
        print ("Reply from", host,
    "Time=", round ((end-start)*1000,2),
                        ("ms")
    else:
        print ("request timed out")
time.sleep(1)
```

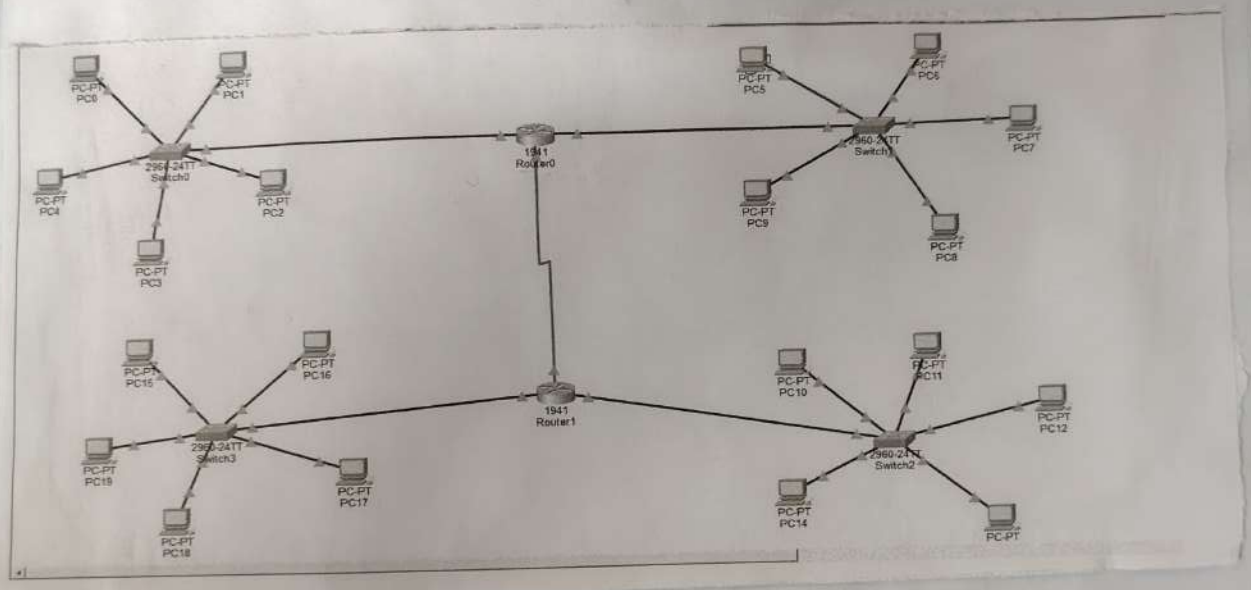
Input:

Enter host: 127.0.0.1

Output:

Reply from 127.0.0.1 Time=1.23ms

Reply from 127.0.0.1 Time=0.98ms.



Result:

The ping program was successfully studied and implemented.

Exp. No: 12
Date: 29/9

Packet Sniffing using raw socket

Aim:

To write a program using RAW sockets to capture and display network packets.

Algorithm:

1. Create a RAW socket.
2. Bind the socket to the network interface.
3. Recieve packets from the network
4. Extract the IP header information
5. Display the source IP, destination IP and protocal.
6. Continue capturing packets.

Code:

```
import socket
s = socket.socket(socket.AF_INET, socket.
    sock_RAW, socket.IPPROTO_IP)
s.bind(("0.0.0.0", 0))
s.setsocket(socket.IPPROTO_IP, socket.
        IP_HDRINCL, 1)
Print("Packet sniffer started...")
While True:
    data, addr = s.recvfrom(65535)
    print("Packet received from:",
                addr[0].
```

Input:

Network packets

Output:

packet sniffer started...

packet received from : 192.168.1.5

packet received from : 192.168.1.1

packet received from : 192.120.183.74.

Result:

The RAW SOCKET SNIFFING PROGRAM was successfully implemented and network packets was captured and display.

Exp. No : 13

Date : 5/9/24 Routing at Network layer.

a) Simulate static routing protocal configuration using cisco packet tracer

Aim :

To simulate a static routing configure using cisco packet tracer.

Procedure :

* Open cisco packet tracer and centre the given network topology.
* connect 3 routers and PCs using suitable cables.
* configure IP address for routers and pcs.
* Enable all router interface using no shutdown
* configure static routers on Router 0, Router 1 and Router 2 using IP mobile.
* Use Tracer to verify the routing path.
* Disconnect, the main link and verify that the backup route works.
* Thus a static routing is successfully configured and verified.

```
Router>enable
Router#show ip route static
S     30.0.0.0/8 [20/0] via 40.0.0.2
S     50.0.0.0/8 [10/0] via 40.0.0.2
S     60.0.0.0/8 [1/0] via 40.0.0.2

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#show ip route static
S     30.0.0.0/8 [10/0] via 20.0.0.2
S     50.0.0.0/8 [10/0] via 40.0.0.2
S     60.0.0.0/8 [1/0] via 40.0.0.2
```

Result:

Thus, static routing was successfully configured cisco packet tracer. connectivity between all networks verified successfully using ping.

6. Simulate RIP using Cisco packet tracer

Aim:

To simulate RIP using Cisco packet tracer

Procedure:

* Create the topology with routers and 2pcs.
* connect the device using suitable cables
* Assign the given IP addresses to both PCs.
* configure Fastethernet and serial interface of all routers.
* Enable interface using no shutdown.
* Let clock rate and 4000 and bandwidth 64 on DE serial interface.
* Enable RIP on all router 0-Router & link and use tracer again to verify RIP selects the alternate route.

```
Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=55ms TTL=125
Reply from 20.0.0.2: bytes=32 time=56ms TTL=125
Reply from 20.0.0.2: bytes=32 time=60ms TTL=125
Reply from 20.0.0.2: bytes=32 time=55ms TTL=125

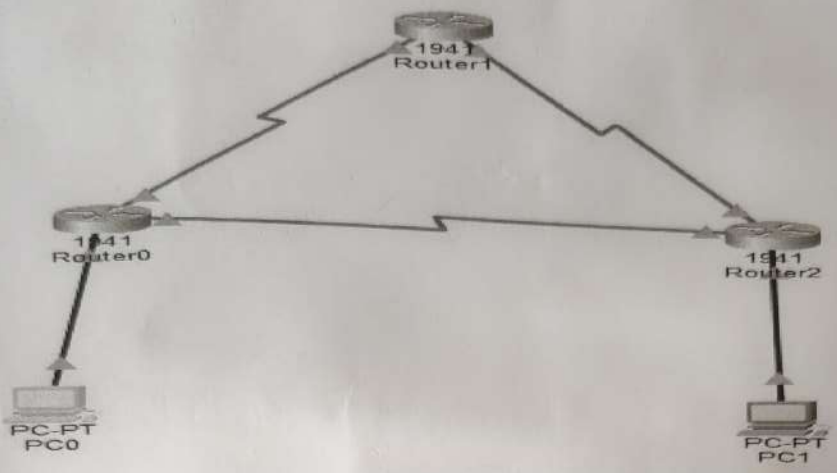
Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 55ms, Maximum = 60ms, Average = 56ms

C:\>tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    11 ms     8 ms      5 ms      192.168.1.250
  3    29 ms     26 ms     25 ms     192.168.1.245
  4    22 ms     28 ms     20 ms     20.0.0.2

Trace complete.
```



Results:

RIP successfully configured and dynamic routing between PC0 and PC1 was verified.

Exp. NO: 14

Date: 6/9/25

INTERWORKING WITH ROUTERS CISCO PACKET TRACER.

a) Design and configure a simple internetwork using a router in cisco packet tracer and fast connectivity between two pcs.

AIM:

To design and configure a sample internetwork using a router in cisco packet tracer and fast connectivity between two pcs.

Procedure:

step 1: configure Router:

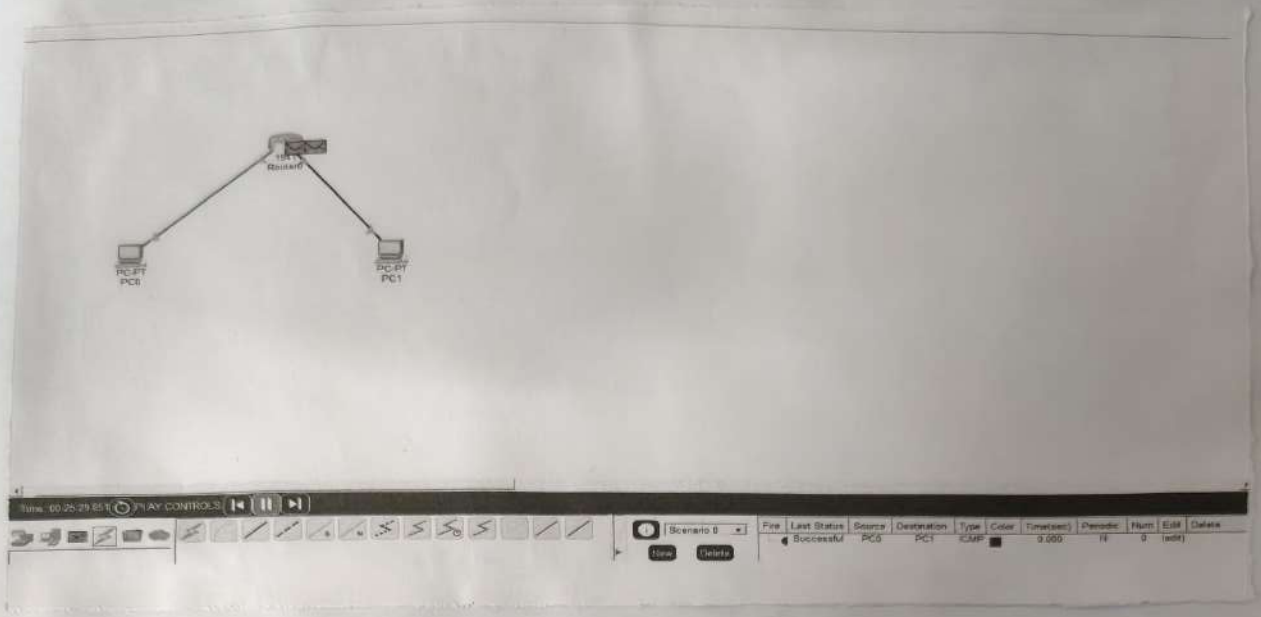
* select the router and open
* press enter to active and unprivileged mode.

step 2:

* Assign IP address to every PC in the network.
* select the PCs, Go to the desktop and select IP configuration and assign an IP address, default gateway subnet mask.
* Assign the default gateway of PC0 as 192.168.10.1
* Assign the default gateway of PC1 as 192.168.20.1

step 3:

* connect fast Ethernet 0 port of PC1 with fast Ethernet 1 Port of Router 1 using a copper straight through cable.



Result:

The simple internetwork was successfully and configured using a router connectivity between PC0 and PC1 was successfully verified using PDU.

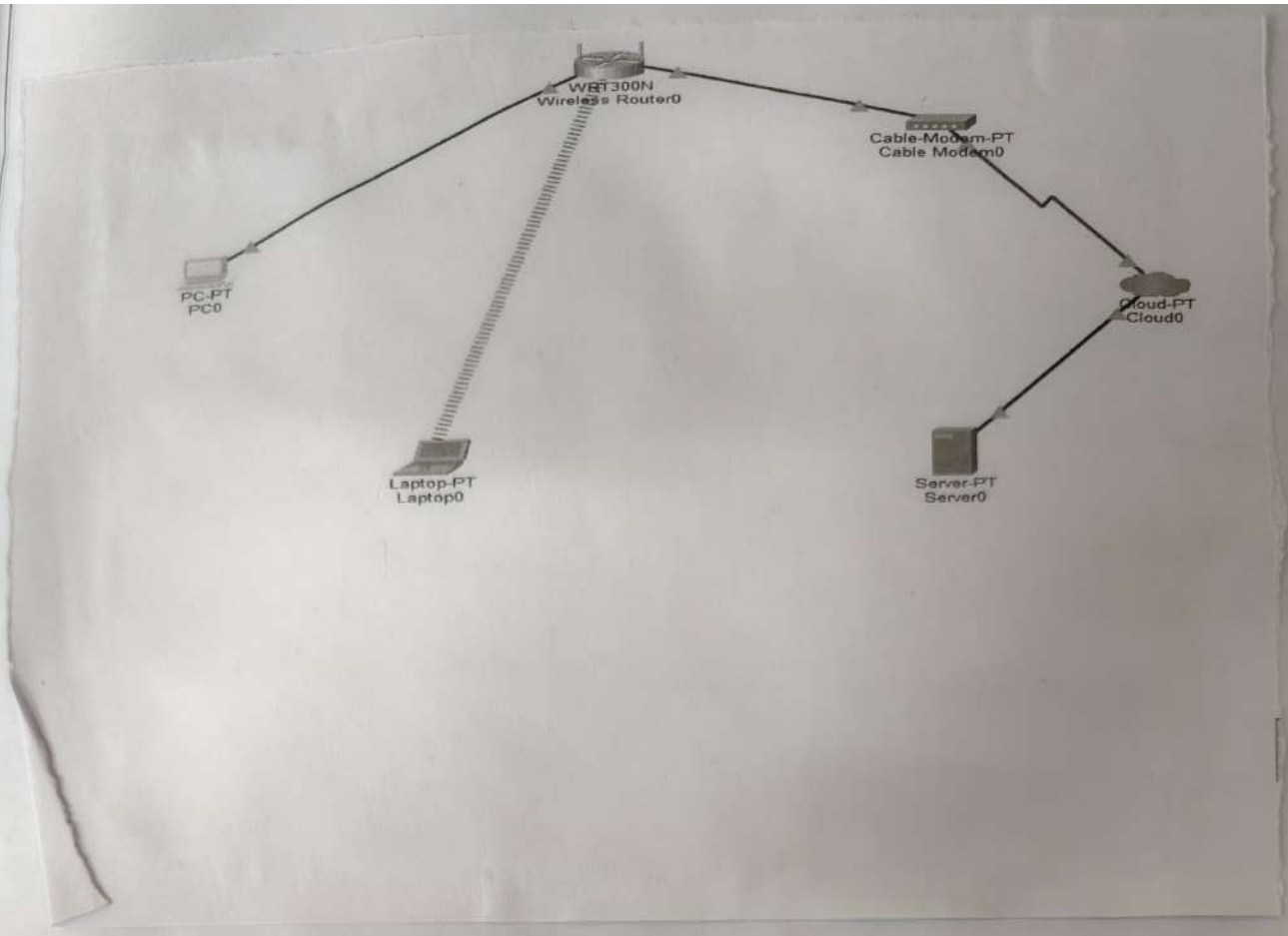
b. configure an internwork using wireless router DHCP server and internet cloud.

AIM:

TO Design and configure and internetwork using a wireless router, DHCP server and internwork cloud in cisco packet tracer.

Procedure:

* open cisco packet tracer and create the required network topology.
* connect PC, laptop, wireless router, cable modem, internet cloud and server.
* configure the wireless router with SSCO Homenetwork and enable DHCP.
* connect the laptop to the wireless network and set the PC to DHCP
* configure the wireless router with IP 208.67. 220. 220.
* Verify connectivity using IP config and ping cisco.com.



Result:

The internetwork was successfully designed and configured using a wireless router, DHCP, server and internet cloud and connectivity was successfully verified using ping. cisco.com.

c. Designed and configuse an intelli - network in your using switch, router and Ethernet cables

AIM:

Design and configused and inetwork using a switch, router and ethernet cables and verify connectivity between computers.

Procedure:

* open cisco packet tracer and place computer, a switch and router in the workplace.

* connect the computer to the switch using ethernet cables.

* connect the switch to the router using an ethernet cable.

* configure the routers interface with appropriate IP address and enable the interface.

* Use the ping command to test connectivity between computers.

* Observe the successfully transmission of packets.

Result:

Thus, an internetwork was successfully designed and configured using a switch, router and ethernet cables in cisco packet tracer.